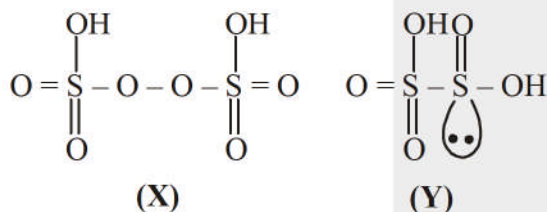
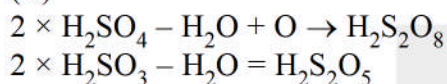


HINTS & SOLUTIONS

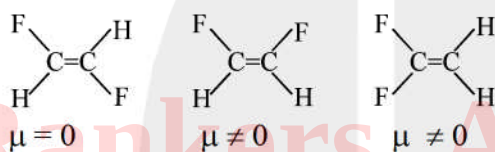
Q.1 (C)



Q.2 (C)

Q.3 (A)

Q.4 (B)



Q.5 (B)

O_2 = Paramagnetic molecule with 2 unpaired electrons in antibonding π -orbital

O_3 = diamagnetic molecule

B_2 = Paramagnetic with 2 unpaired electron in bonding π -orbitals

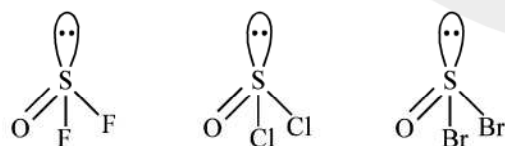
C_2 = Diamagnetic molecule

$\text{NO} \longrightarrow \text{NO}^+$ (Electron removes from antibonding orbital $\therefore$ bond length decreases)

$\text{CO} \longrightarrow \text{CO}^+$ (Electron removes from antibonding orbital $\therefore$ bond length decreases)]

Q.6 (C)

Q.7 (B)



In SOBr_2 , S–O bond has maximum bond length in comparison to S–O bond lengths in SOF_2 and SOCl_2 , because in SOBr_2 , S–O bond has been formed by hybrid orbital containing less s-character. According to Bent's Rule

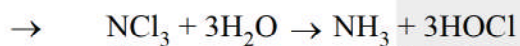
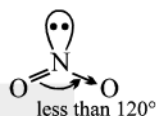
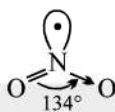
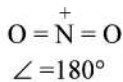
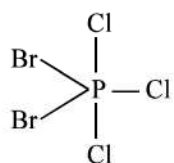
Q.8 (C)

Q.9 (B)

Q.10 (C)

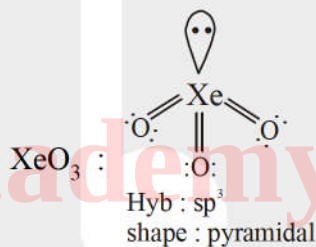
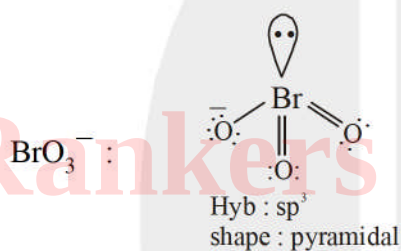
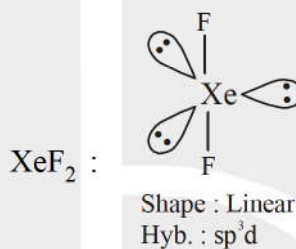
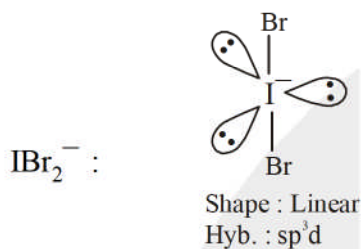
Q.11 (A)

Q.12 (D)



→ Silicones are water repellant due to presence of hydrophobic groups

Q.13 (A)

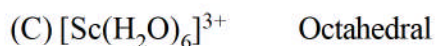
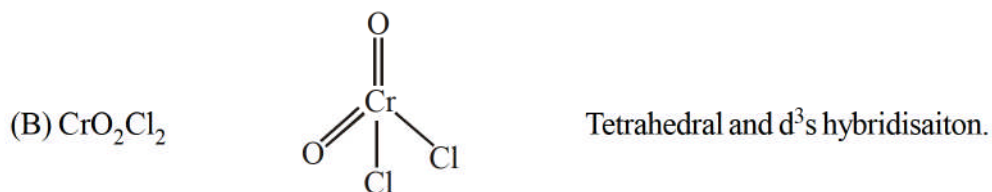
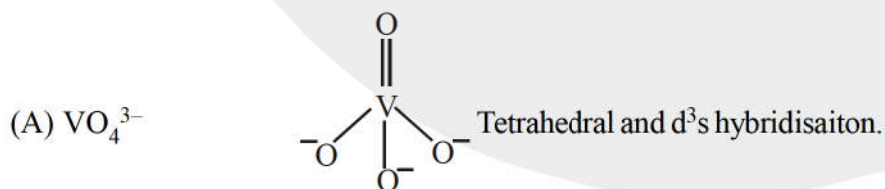


Q.14 (D)

Q.15 (C)

CO_2 molecule always exist gases form at room temperature

Q.16 (D)

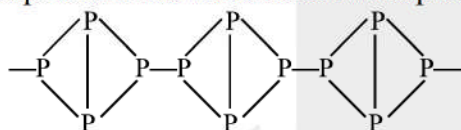


- Q.17 (B)
 (A) $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4$
 (B) $\text{Xe} > \text{Ne} > \text{He}$
 (C) $\text{H}_2 > \text{He}$
 (D) $\text{CH}_4 < \text{CD}_4$

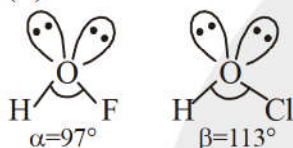
- Q.18 (C)
 As the difference in electronegativity of HX increases, dipole moment increases.

- Q.19 (D)

- Q.20 (C)
 Red phosphorus is less volatile than white phosphorus because it exists in linked tetrahedral structures

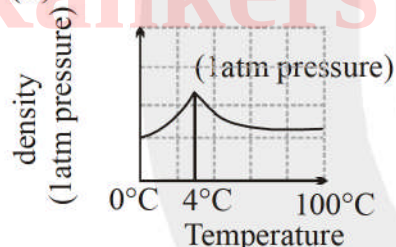


- Q.21 (B)



In O-F bond p-character is greater than that of O-Cl bond, also in HOCl, $\pi\pi$ - $d\pi$ back bonding may also be present, hence $\alpha < \beta$.

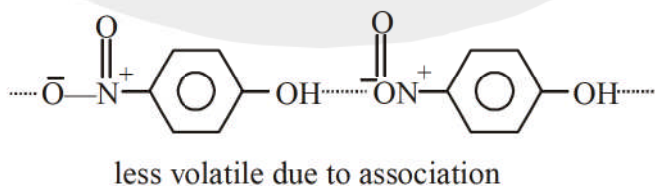
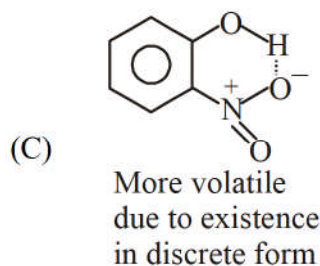
- Q.22 (A)

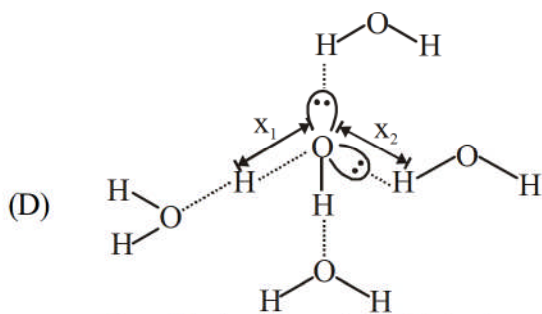


Ice has less density than water at all temperature but below boiling point.

- Q.23 (D)

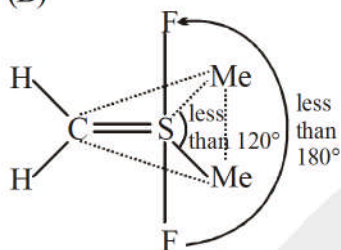
- (A) In H_2O_2 , extent of H-bonding is greater than that of water.
 (B) Glycerol is highly viscous than ethylene glycol because in glycerol per molecule number of -OH groups are three while in ethylene glycol they are two which can form intermolecular H-bond.





Four H-atom are not equidistant
 $x_2 > x_1$

Q.24 (B)

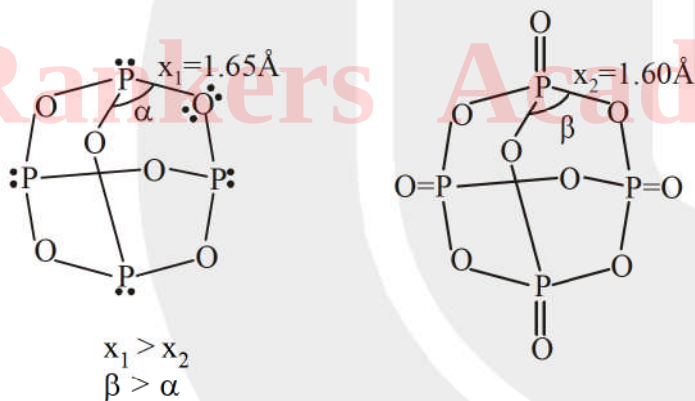


Hybridisation : sp^3d

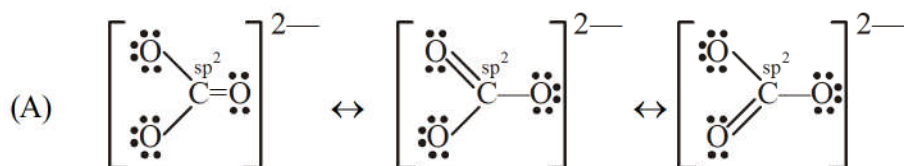
Both C-H bonds are in axial plane ;
 π bond is formed by $p\pi - d\pi$ overlapping.

Nodal plane of π bond lies in axial plane

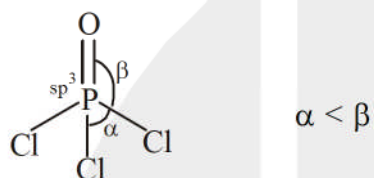
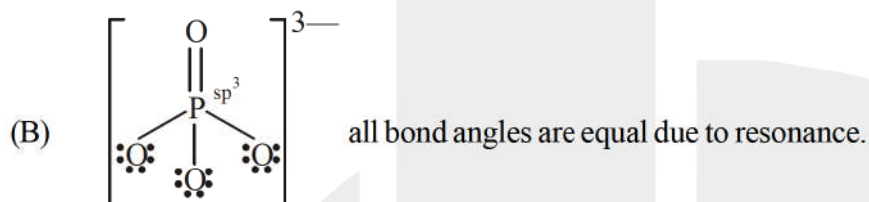
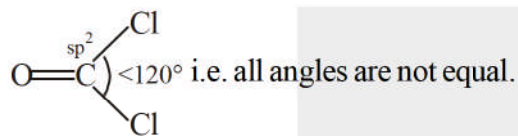
Q.25 (B)



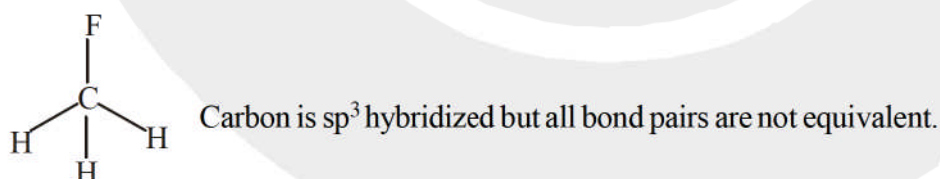
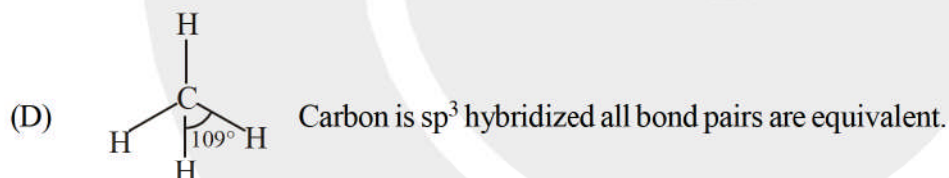
Q.26 (C)



Due to resonance all bond angles are equal in CO_3^{2-}



All bond angles are equal because % of s and p character are same in each bond pairs of both molecules.



Hence all bond angles are not equal in this pair.

Q.27 (A)

Q.28 (B)

Q.29 (A)

Q.30 (B)

Q.31 (B)

⇒ As DCl_3 is trigonal planar, hence 'D' belongs to Boron family.

⇒ As AlCl_3 is trigonal pyramidal, hence 'A' belongs to Nitrogen family.

⇒ $\text{DCl}_3 + \text{AlCl}_3 \longrightarrow [\text{DCl}_2]^+ [\text{AlCl}_4]^-$ or $[\text{AlCl}_2]^+ [\text{DCl}_4]^-$

If anionic part is $[\text{AlCl}_4]^-$ which has Sea-saw structure then cationic part will be $[\text{DCl}_2]^+$ where Hybridization of D will be 'sp' and hence shape will be linear.

⇒ If anionic part is DCl_4^- i.e. it is of tetrahedral shape then cation part will be AlCl_2^+ which will be bent]

Q.32 (B) (D)

X is CO_2 and Y is CO

Q.33 (A) (B) (C) (D)

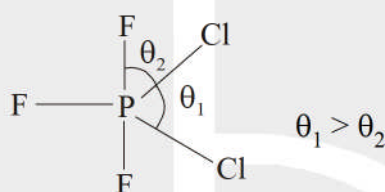
Suppose

X & Y are F and Cl respectively

$\text{PX}_n\text{Y}_{5-n}$

If $n = 3$

PF_3Cl_2

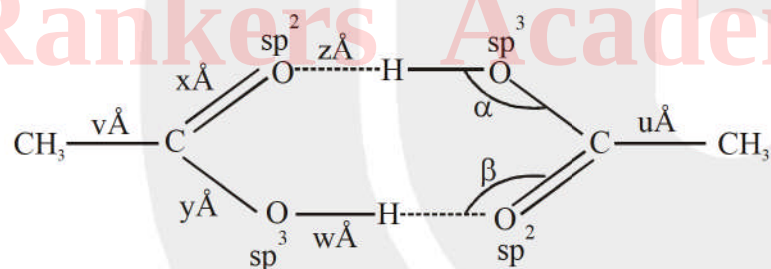


If $n = 0$

PCl_5

Due to the presence of different bond angle it having different bond length.

Q.34 (A) (B) (C) (D)



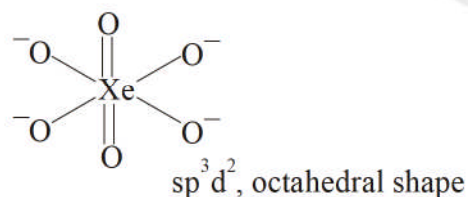
(A) Bond order increase bond length decrease

(B) Covalent bond is stronger than hydrogen bond, due to the more attraction it is shorter than H-bond.

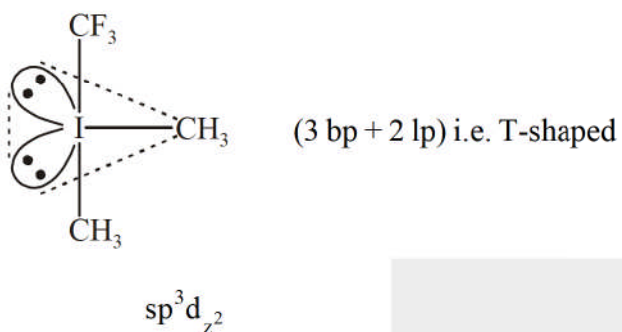
(C) Multiple bond single bond repulsion is greater than single-single bond repulsion

(D) C-C bond remain unaffected almost

Q.35 (A) (B) (C) (D)



Q.36 (A) (B) (C) (D)

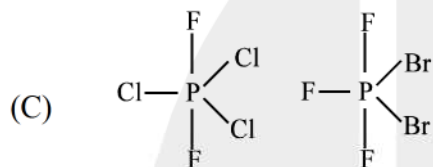


Q.37 (A) (B) (C)

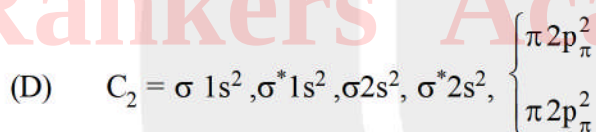


(Hydrogen peroxy phosphate)

(B) Both PH_5 and NH_5 do not exist

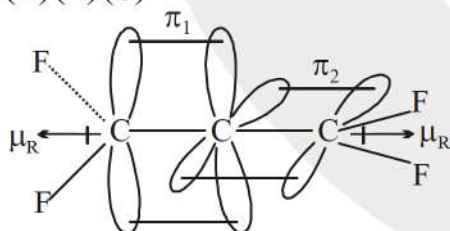


$\angle C/PCl$ and $\angle BrPBr$ are not equal



B.O. = 2, all stabilized electron present in π bonding M.O. so there are two covalent bonds between two C atoms in which both are π bonds.

Q.38 (A) (B) (C)



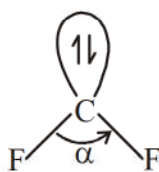
Non planar

$\mu = 0$ i.e. Non planar

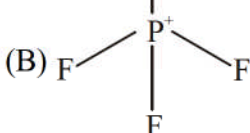
Nodal planes of π_1 & π_2 are perpendicular to each other.

Q.39 (A) (B) (D)

(A) :CF_2 It may be either singlet or triplet form

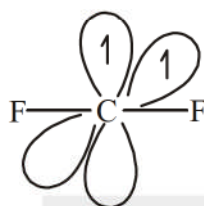
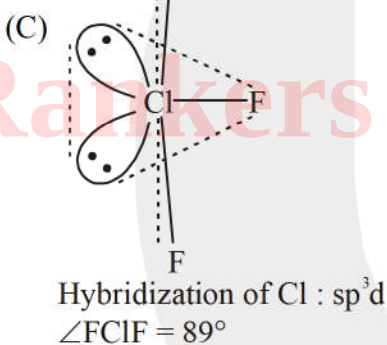


Singlet
Hybridization of C : sp^2
 $\alpha \approx 120^\circ$

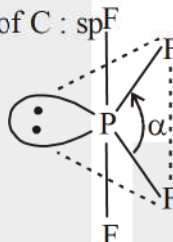


Hybridization of P : sp^3
 $\angle = 109^\circ 28'$

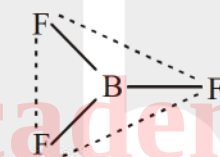
(D) In NH_3 , $\angle \text{H-N-H} = 107^\circ$
In PH_3 , $\angle \text{H-P-H} = 93.8^\circ$



Triplet
Hybridization of C : sp
 $\angle \text{FCF} = 180^\circ$

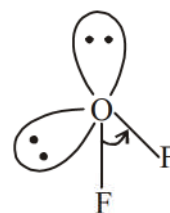


Hybridization of P : sp^3d
 $\angle \text{FPF (eq.)} < 120^\circ$
 $\angle \text{F}_{\text{eq}} \text{P F}_{\text{(axial)}} < 90^\circ$



Hybridization of B : sp^2
 $\angle \text{FBF} = 120^\circ$

OF_2 :

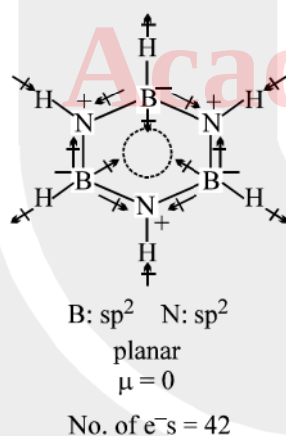
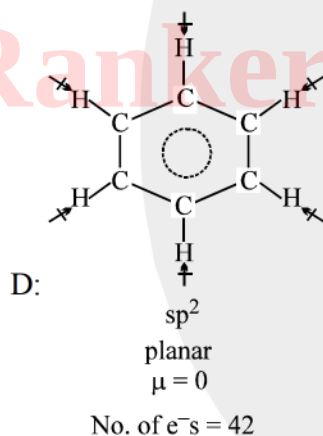
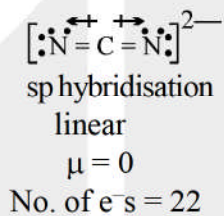
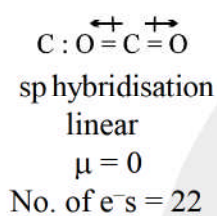
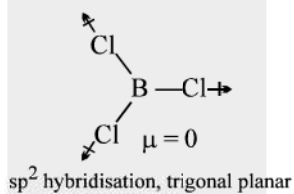
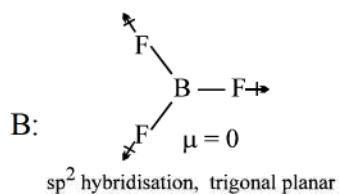
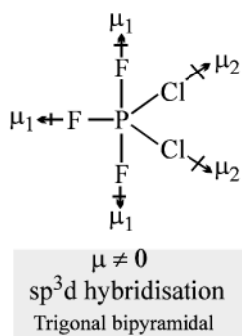
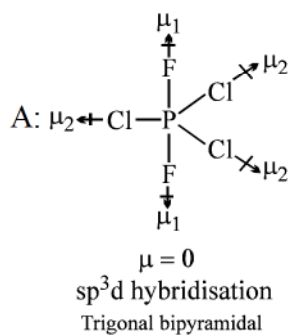


Hybridization of O : sp^3
 $\angle \text{FOF} = 103^\circ$

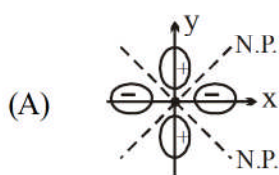
Q.40 (A) R (B) S (C) P (D) Q]

Q.41 (A) R, T ; (B) Q, R ; (C) P, R ; (D) P, R

Q.42 (A) P, Q; (B) P, Q, R; (C) P,Q,R,S ; (D) P,Q,R,S



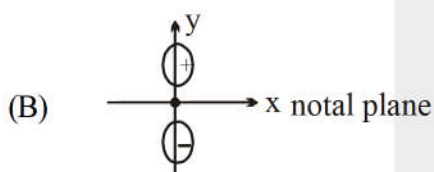
Q.43 (A) Q, R, (B) P, S, (C) R, (D) Q, R



$d_{x^2-y^2}$ orbital

No. of nodal planes = 2

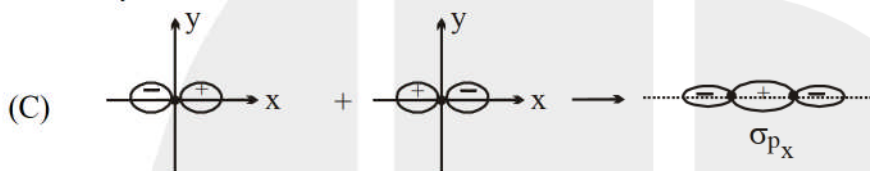
$d_{x^2-y^2}$ has even no. of nodal planes hence it is gerade (i.e. ψ_g)



(p_y orbital)

No. of nodal plane = 1

p_y has odd no. of nodal planes hence it is ungerade (i.e. ψ_u)



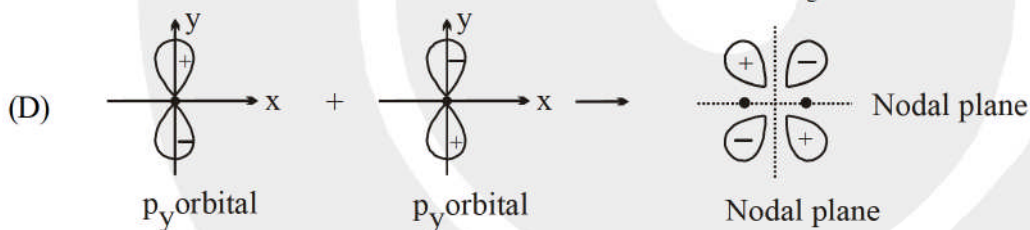
p_x orbital

p_x orbital

σ_{p_x}

No. of nodal plane = 0

M.O. has even no. of nodal planes hence it is gerade (i.e. ψ_g)



p_y orbital

p_y orbital

Nodal plane

Nodal plane

No. of nodal plane = 2

π_{2py}^* has even no. of nodal plane hence it is gerade (i.e. ψ_g)

Q.44 6

Planar molecules

ClF_3 , XeF_2 , XeF_4 , H_2O , H_2S , OCl_2

Q.45 9

NO_3^- , CO_3^{2-} , F_2 , Cl_2 , Br_2 , O_2^{2-} , O_2^- , Li_2^+ , He_2^+

Q.46 5

Q.47 4

It vapour state of HF be $(\text{HF})_n$

$$P \times \text{mol. wt.} = dRT$$

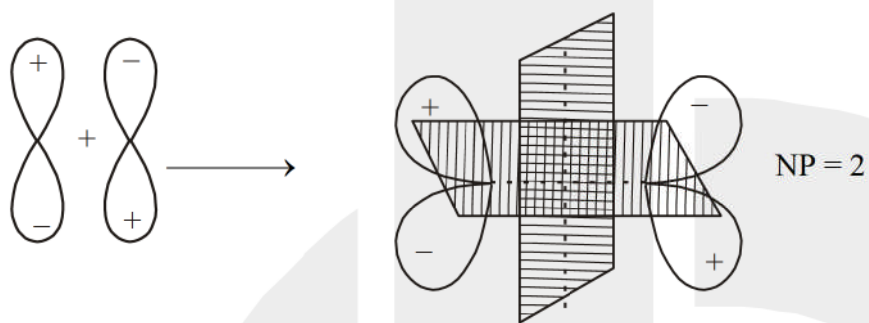
$$1 \times \text{Mol. wt.} = 3.248 \times 0.0821 \times 300$$

$$n(20) = \text{mol. wt.} = 79.9 \simeq 80$$

$$n = 4$$

hence vapour state is represented as $(\text{HF})_4$

Q.48 2



Q.49 4

ClO_2 , NO_2 , SO_2 , $\bullet\text{CH}_3$, N_2H_4 are sp^2 hybridised

